ASSIGNMENT-17.1

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Batch: 36

Task 1 – Customer Data Cleaning

Task:

Use AI to generate a Python script for cleaning a customer dataset.

Instructions:

• Handle missing values in columns (age, income, city).

• Remove duplicate records.

• Standardize text values (e.g., city names in lowercase).

• Encode categorical variables (gender, city).

Sample Code:

```
import pandas as pd
data = pd.read_csv("shopping_trends.csv")
print(data.head())
```

Dataset link:

https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset?select=shopping_trends.csv

Expected Output:

• A cleaned Pandas data frame ready for analysis.

```python
# Scale numerical features using MinMaxScaler
scaler = MinMaxScaler()

# Identify numerical columns to scale (excluding original categorical and 'id')
# Also exclude newly created encoded columns from scaling
all_cols = set(healthcare_df.columns)
encoded_cols = {col for col in healthcare_df.columns if '_Encoded' in col}
original_categorical_cols = set(categorical_cols)

# Numerical columns are those that are not object type, not encoded, and not 'id'
numerical_cols = [col for col in healthcare_df.select_dtypes(include=np.number).columns
                  if col not in encoded_cols and col != 'id']

for col in numerical_cols:
    healthcare_df[col + '_Scaled'] = scaler.fit_transform(healthcare_df[[col]])

print("\nDataFrame Head after scaling numerical features:")
display(healthcare_df.head())
```

```
DataFrame Head after scaling numerical features:
```

	Age	Gender	Medical Condition	Glucose	Blood Pressure	BMI	Oxygen Saturation	LengthOfStay	Cholesterol	Triglycerides	...	Triglycerides_Scaled	HbA1c_Scale
0	48.0	Male	Diabetes	137.04	135.27	28.90	96.04	6	231.88	210.56	...	0.524877	0.47687
1	22.0	Male	Healthy	71.58	113.27	26.29	97.54	2	165.57	129.41	...	0.342102	0.17951
2	50.0	NaN	Asthma	95.24	138.32	22.53	90.31	2	214.94	169.36	...	0.423050	0.28590
3	57.0	NaN	Obesity	NaN	130.53	38.47	96.80	6	187.71	192.13	...	0.493844	0.40388

```
#  Color                   3900 non-null   object
9  Season                  3900 non-null   object
10 Review Rating           3900 non-null   float64
11 Subscription Status     3900 non-null   object
12 Payment Method          3900 non-null   object
13 Shipping Type           3900 non-null   object
14 Discount Applied        3900 non-null   object
15 Promo Code Used         3900 non-null   object
16 Previous Purchases      3900 non-null   int64
17 Preferred Payment Method 3900 non-null  object
18 Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(14)
memory usage: 579.8+ KB

Original DataFrame Head:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Payment Method	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Preferred Payment Method	Fr
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Credit Card	Express	Yes	Yes	14	Venmo	F
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Bank Transfer	Express	Yes	Yes	2	Cash	F
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Cash	Free Shipping	Yes	Yes	23	Credit Card	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	PayPal	Next Day Air	Yes	Yes	49	PayPal	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Cash	Free Shipping	Yes	Yes	31	PayPal	

Now, let's proceed with cleaning the dataset. We'll start by handling missing values in the specified columns.

```python
# Handle missing values
# For numerical columns ('Age', 'Purchase Amount (USD)'), we'll impute with the median.
# For categorical columns ('Gender', 'Item Purchased', 'Category', 'City', 'Season', 'Subscription Status', 'Payment Method', 'Shipping Type', 'Discount Applied', 'Promo Cod

# Age: Impute with median
df['Age'].fillna(df['Age'].median(), inplace=True)

# Income (assuming 'Purchase Amount (USD)' is the closest to 'Income' for this dataset, or 'Age' as no income is present): No 'Income' column, using 'Purchase Amount (USD)'
# If 'Income' was explicitly present, the approach would be similar to 'Age'.
df['Purchase Amount (USD)'].fillna(df['Purchase Amount (USD)'].median(), inplace=True)

# City: Impute with mode
df['Location'].fillna(df['Location'].mode()[0], inplace=True)

print("\nDataFrame Info after handling missing values:")
df.info()

print("\nNumber of missing values after imputation:")
print(df.isnull().sum())
```

```
DataFrame Info after handling missing values:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 19 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
```

```
 10  Review Rating           3900 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Payment Method          3900 non-null   object
 13  Shipping Type           3900 non-null   object
 14  Discount Applied        3900 non-null   object
 15  Promo Code Used         3900 non-null   object
 16  Previous Purchases      3900 non-null   int64
 17  Preferred Payment Method 3900 non-null  object
 18  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(14)
memory usage: 570.0+ KB

Number of missing values after imputation:
Customer ID                 0
Age                         0
Gender                      0
Item Purchased              0
Category                    0
Purchase Amount (USD)       0
Location                    0
Size                        0
Color                       0
Season                      0
Review Rating               0
Subscription Status         0
Payment Method              0
Shipping Type               0
Discount Applied            0
Promo Code Used             0
Previous Purchases          0
Preferred Payment Method    0
Frequency of Purchases      0
dtype: int64
/tmp/ipykernel_9607/97586187.py:6: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.
```

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Next, we will remove any duplicate records from the dataset.

[6]

```python
# Remove duplicate records
initial_rows = df.shape[0]
df.drop_duplicates(inplace=True)
final_rows = df.shape[0]

print(f"Number of rows before removing duplicates: {initial_rows}")
print(f"Number of rows after removing duplicates: {final_rows}")
print(f"Number of duplicate records removed: {initial_rows - final_rows}")
```

```
Number of rows before removing duplicates: 3900
Number of rows after removing duplicates: 3900
Number of duplicate records removed: 0
```

Now, let's standardize text values, specifically by converting city names to lowercase.

[7]

```python
# Standardize text values (e.g., city names in lowercase)
df['Location'] = df['Location'].str.lower()

print("\nUnique city names after standardization:")
print(df['Location'].unique()[:10]) # Display first 10 unique cities
```

```
Unique city names after standardization:
['kentucky' 'maine' 'massachusetts' 'rhode island' 'oregon' 'wyoming'
 'montana' 'louisiana' 'west virginia' 'missouri']
```

Finally, we will encode categorical variables ('Gender' and 'Location'). We'll use `LabelEncoder` for 'Gender' and 'Location' to convert them into numerical representations, which is suitable for many machine learning models.

```python
from sklearn.preprocessing import LabelEncoder

# Encode categorical variables
le = LabelEncoder()

# Encode 'Gender'
df['Gender_Encoded'] = le.fit_transform(df['Gender'])
print(f"Original Gender categories: {le.classes_}")

# Encode 'Location'
df['Location_Encoded'] = le.fit_transform(df['Location'])
print(f"Original Location categories (first 10): {le.classes_[:10]}...")

print("\nDataFrame head after encoding categorical variables:")
display(df.head())
```

```
Original Gender categories: ['Female' 'Male']
Original Location categories (first 10): ['alabama' 'alaska' 'arizona' 'arkansas' 'california' 'colorado'
 'connecticut' 'delaware' 'florida' 'georgia']...

DataFrame head after encoding categorical variables:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	...	Subscription Status	Payment Method	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Preferred Payment Method	Freq Purc
0	1	55	Male	Blouse	Clothing	53	kentucky	L	Gray	Winter	...	Yes	Credit Card	Express	Yes	Yes	14	Venmo	Fort
1	2	19	Male	Sweater	Clothing	64	maine	L	Maroon	Winter	...	Yes	Bank Transfer	Express	Yes	Yes	2	Cash	Fort
2	3	50	Male	Jeans	Clothing	73	massachusetts	S	Maroon	Spring	...	Yes	Cash	Free Shipping	Yes	Yes	23	Credit Card	W

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```python
print("Final Cleaned DataFrame Info:")
df.info()

print("\nFinal Cleaned DataFrame Head:")
display(df.head())
```

```
Final Cleaned DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 21 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Customer ID               3900 non-null   int64
 1   Age                       3900 non-null   int64
 2   Gender                    3900 non-null   object
 3   Item Purchased            3900 non-null   object
 4   Category                  3900 non-null   object
 5   Purchase Amount (USD)     3900 non-null   int64
 6   Location                  3900 non-null   object
 7   Size                      3900 non-null   object
 8   Color                     3900 non-null   object
 9   Season                    3900 non-null   object
 10  Review Rating             3900 non-null   float64
 11  Subscription Status       3900 non-null   object
 12  Payment Method            3900 non-null   object
 13  Shipping Type             3900 non-null   object
 14  Discount Applied          3900 non-null   object
 15  Promo Code Used           3900 non-null   object
 16  Previous Purchases        3900 non-null   int64
 17  Preferred Payment Method  3900 non-null   object
 18  Frequency of Purchases    3900 non-null   object
 19  Gender_Encoded            3900 non-null   int64
 20  Location_Encoded          3900 non-null   int64
```

```
 12  Payment Method            3900 non-null   object
 13  Shipping Type             3900 non-null   object
 14  Discount Applied          3900 non-null   object
 15  Promo Code Used           3900 non-null   object
 16  Previous Purchases        3900 non-null   int64
 17  Preferred Payment Method  3900 non-null   object
 18  Frequency of Purchases    3900 non-null   object
 19  Gender_Encoded            3900 non-null   int64
 20  Location_Encoded          3900 non-null   int64
dtypes: float64(1), int64(6), object(14)
memory usage: 640.0+ KB

Final Cleaned DataFrame Head:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	...	Subscription Status	Payment Method	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Preferred Payment Method	Freq Purcl
0	1	55	Male	Blouse	Clothing	53	kentucky	L	Gray	Winter	...	Yes	Credit Card	Express	Yes	Yes	14	Venmo	Fortr
1	2	19	Male	Sweater	Clothing	64	maine	L	Maroon	Winter	...	Yes	Bank Transfer	Express	Yes	Yes	2	Cash	Fortr
2	3	50	Male	Jeans	Clothing	73	massachusetts	S	Maroon	Spring	...	Yes	Cash	Free Shipping	Yes	Yes	23	Credit Card	W
3	4	21	Male	Sandals	Footwear	90	rhode island	M	Maroon	Spring	...	Yes	PayPal	Next Day Air	Yes	Yes	49	PayPal	W
4	5	45	Male	Blouse	Clothing	49	oregon	M	Turquoise	Spring	...	Yes	Cash	Free Shipping	Yes	Yes	31	PayPal	An

5 rows × 21 columns

Task 2 – Sales Data Preprocessing

Task:

Preprocess sales transaction data using AI assistance.

Instructions:

• Convert date columns into datetime format.

• Create new features (month, year, day of week).

• Normalize sales amount column.

• Detect and handle outliers in dataset.

Sample Code:

```
import pandas as pd
data = pd.read_csv("sales_data.csv")
print(data.head())
```

Dataset link:

https://www.kaggle.com/datasets/vinothkannaece/sales-dataset?select=sales_data.csv

Expected Output:

• A transformed dataset with time-based features and normalized values.

```python
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
import numpy as np

# Load the sales dataset. Assuming it's uploaded to the Colab environment.
# If using a remote URL, replace '/content/sales_data.csv' with the direct URL.
data_url = "/content/sales_data.csv"
sales_df = pd.read_csv(data_url)

print("Original Sales DataFrame Info:")
sales_df.info()

print("\nOriginal Sales DataFrame Head:")
display(sales_df.head())
```

```
Original Sales DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Product_ID            1000 non-null   int64
 1   Sale_Date             1000 non-null   object
 2   Sales_Rep             1000 non-null   object
 3   Region                1000 non-null   object
 4   Sales_Amount          1000 non-null   float64
 5   Quantity_Sold         1000 non-null   int64
 6   Product_Category      1000 non-null   object
 7   Unit_Cost             1000 non-null   float64
 8   Unit_Price            1000 non-null   float64
 9   Customer_Type         1000 non-null   object
 10  Discount              1000 non-null   float64
 11  Payment_Method        1000 non-null   object
```

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```python
# Convert 'Date' column to datetime format
sales_df['Sale_Date'] = pd.to_datetime(sales_df['Sale_Date'])

# Create new features: month, year, day of week
sales_df['Month'] = sales_df['Sale_Date'].dt.month
sales_df['Year'] = sales_df['Sale_Date'].dt.year
sales_df['DayOfWeek'] = sales_df['Sale_Date'].dt.dayofweek # Monday=0, Sunday=6

print("\nDataFrame Head with new date features:")
display(sales_df.head())
```

DataFrame Head with new date features:

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Regio
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	
2	1015	2023-08-21	David	South	4631.23	30	Food	261.50	371.40	Returning	0.20	Bank Transfer	Retail	
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	
4	1001	2023-03-24	Charlie	East	3790.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	

deletion use Ctrl+M Z or the Undo option in the Edit menu

```python
# Normalize 'Sales_Amount' column
scaler = MinMaxScaler()
sales_df['Sales_Amount_Normalized'] = scaler.fit_transform(sales_df[['Sales_Amount']])

print("\nDataFrame Head with normalized sales amount:")
display(sales_df.head())
```

```
DataFrame Head with normalized sales amount:
```

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Regi
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	
2	1015	2023-09-21	David	South	4831.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.06	Credit Card	Online	

Finally, we will detect and handle outliers in the 'Sales_Amount' column using the Interquartile Range (IQR) method. Outliers will be capped.

```python
# Detect and handle outliers in 'Sales_Amount' using IQR
Q1 = sales_df['Sales_Amount'].quantile(0.25)
Q3 = sales_df['Sales_Amount'].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

sales_df['Sales_Amount_Capped'] = sales_df['Sales_Amount'].clip(lower=lower_bound, upper=upper_bound)

print(f"\nOutlier Detection (IQR Method) for Sales_Amount:")
print(f"Q1: {Q1}, Q3: {Q3}, IQR: {IQR}")
print(f"Lower Bound: {lower_bound}, Upper Bound: {upper_bound}")
print(f"Number of outliers below lower bound: {(sales_df['Sales_Amount'] < lower_bound).sum()}")
print(f"Number of outliers above upper bound: {(sales_df['Sales_Amount'] > upper_bound).sum()}")

print("\nDataFrame Head with capped sales amount:")
display(sales_df.head())
```

```
Outlier Detection (IQR Method) for Sales_Amount:
Q1: 2550.2975, Q3: 7507.445000000001, IQR: 4957.147500000001
Lower Bound: -4885.423750000002, Upper Bound: 14941.166250000002
Number of outliers below lower bound: 0
Number of outliers above upper bound: 0

DataFrame Head with capped sales amount:
```

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Regi
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	

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```python
print("Final Preprocessed Sales DataFrame Info:")
sales_df.info()

print("\nFinal Preprocessed Sales DataFrame Head:")
display(sales_df.head())
```

```
Final Preprocessed Sales DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 19 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Product_ID               1000 non-null   int64
 1   Sale_Date                1000 non-null   datetime64[ns]
 2   Sales_Rep                1000 non-null   object
 3   Region                   1000 non-null   object
 4   Sales_Amount             1000 non-null   float64
 5   Quantity_Sold            1000 non-null   int64
 6   Product_Category         1000 non-null   object
 7   Unit_Cost                1000 non-null   float64
 8   Unit_Price               1000 non-null   float64
 9   Customer_Type            1000 non-null   object
 10  Discount                 1000 non-null   float64
 11  Payment_Method           1000 non-null   object
 12  Sales_Channel            1000 non-null   object
 13  Region_and_Sales_Rep     1000 non-null   object
 14  Month                    1000 non-null   int32
 15  Year                     1000 non-null   int32
 16  DayOfWeek                1000 non-null   int32
 17  Sales_Amount_Normalized  1000 non-null   float64
 18  Sales_Amount_Capped      1000 non-null   float64
dtypes: datetime64[ns](1), float64(6), int32(3), int64(2), object(7)
memory usage: 136.8+ KB
```

Task 3 – Healthcare Data Preparation

Task:

Clean and preprocess a healthcare dataset using AI scripts.

Instructions:

• Handle missing values in blood_pressure, cholesterol.

• Encode categorical features

• Scale numerical features (min-max or standard scaling).

• Split dataset into training and testing sets.

Dataset link :

https://www.kaggle.com/datasets/abdallaahmed77/healthcare-risk-factors-dataset

Expected Output:

• A preprocessed dataset ready for machine learning models.

```python
import pandas as pd
from sklearn.preprocessing import LabelEncoder, MinMaxScaler
from sklearn.model_selection import train_test_split

# Load the healthcare dataset from the provided link
data_url = "/content/dirty_v3_path.csv"
healthcare_df = pd.read_csv(data_url)

print("Original Healthcare DataFrame Info:")
healthcare_df.info()

print("\nOriginal Healthcare DataFrame Head:")
display(healthcare_df.head())
```

```
Original Healthcare DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 20 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Age                25500 non-null  float64
 1   Gender             25500 non-null  object
 2   Medical Condition  25500 non-null  object
 3   Glucose            25500 non-null  float64
 4   Blood Pressure     25500 non-null  float64
 5   BMI                30000 non-null  float64
 6   Oxygen Saturation  30000 non-null  float64
 7   LengthOfStay       30000 non-null  int64
 8   Cholesterol        30000 non-null  float64
 9   Triglycerides      30000 non-null  float64
 10  HbA1c              30000 non-null  float64
 11  Smoking            30000 non-null  int64
 12  Alcohol            30000 non-null  int64
 13  Physical Activity  30000 non-null  float64
```

```
memory usage: 8.6+ MB
```

Original Healthcare DataFrame Head:

	Age	Gender	Medical Condition	Glucose	Blood Pressure	BMI	Oxygen Saturation	LengthOfStay	Cholesterol	Triglycerides	HbA1c	Smoking	Alcohol	Physical Activity	Diet Score	Family History	Stress Level	Sleep Hours	re
0	46.0	Male	Diabetes	137.04	135.27	28.90	98.04	6	231.68	210.56	7.61	0	0	-0.20	3.54	0	5.07	6.05	
1	22.0	Male	Healthy	71.58	113.27	26.29	97.54	2	165.57	129.41	4.91	0	0	8.12	5.90	0	5.87	7.72	
2	50.0	NaN	Asthma	95.24	NaN	22.53	90.31	2	214.94	165.38	5.80	0	0	5.01	4.85	1	3.09	4.82	
3	57.0	NaN	Obesity	NaN	130.53	38.47	96.60	5	197.71	182.13	6.32	0	0	3.16	3.37	0	3.01	5.33	
4	66.0	Female	Hypertension	95.15	178.17	31.12	94.90	4	269.53	115.85	5.98	0	1	5.56	3.40	0	6.38	6.64	

Task 3: Healthcare Data Preparation

Let's start by handling missing values in the `blood_pressure` and `cholesterol` columns. For numerical features, median imputation is a robust strategy.

```
# Handle missing values in 'blood_pressure' and 'cholesterol'
# Using median imputation for numerical features
healthcare_df['Blood Pressure'] = healthcare_df['Blood Pressure'].fillna(healthcare_df['Blood Pressure'].median())
healthcare_df['Cholesterol'] = healthcare_df['Cholesterol'].fillna(healthcare_df['Cholesterol'].median())

print("\nHealthcare DataFrame Info after handling missing values:")
healthcare_df.info()

print("\nNumber of missing values after imputation:")
```

```
Healthcare DataFrame Info after handling missing values:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 40 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Age                        25500 non-null  float64
 1   Gender                     25500 non-null  object
 2   Medical Condition          25500 non-null  object
 3   Glucose                    25500 non-null  float64
 4   Blood Pressure             30000 non-null  float64
 5   BMI                        30000 non-null  float64
 6   Oxygen Saturation          30000 non-null  float64
 7   LengthOfStay               30000 non-null  int64
 8   Cholesterol                30000 non-null  float64
 9   Triglycerides              30000 non-null  float64
 10  HbA1c                      30000 non-null  float64
 11  Smoking                    30000 non-null  int64
 12  Alcohol                    30000 non-null  int64
 13  Physical Activity          30000 non-null  float64
 14  Diet Score                 30000 non-null  float64
 15  Family History             30000 non-null  int64
 16  Stress Level               30000 non-null  float64
 17  Sleep Hours                30000 non-null  float64
 18  random_notes               30000 non-null  object
 19  noise_col                  30000 non-null  float64
 20  Gender_Encoded             30000 non-null  int64
 21  Medical Condition_Encoded  30000 non-null  int64
 22  random_notes_Encoded       30000 non-null  int64
 23  Age_Scaled                 25500 non-null  float64
 24  Glucose_Scaled             25500 non-null  float64
 25  Blood Pressure_Scaled      30000 non-null  float64
 26  BMI_Scaled                 30000 non-null  float64
 27  Oxygen Saturation_Scaled   30000 non-null  float64
```

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```
Alcohol                          0
Physical Activity                0
Diet Score                       0
Family History                   0
Stress Level                     0
Sleep Hours                      0
random_notes                     0
noise_col                        0
Gender_Encoded                   0
Medical Condition_Encoded        0
random_notes_Encoded             0
Age_Scaled                    4500
Glucose_Scaled                4500
Blood Pressure_Scaled            0
BMI_Scaled                       0
Oxygen Saturation_Scaled         0
LengthOfStay_Scaled              0
Cholesterol_Scaled               0
Triglycerides_Scaled             0
HbA1c_Scaled                     0
Smoking_Scaled                   0
Alcohol_Scaled                   0
Physical Activity_Scaled         0
Diet Score_Scaled                0
Family History_Scaled            0
Stress Level_Scaled              0
Sleep Hours_Scaled               0
noise_col_Scaled                 0
dtype: int64
```

Next, we will encode categorical features. We'll identify object-type columns and apply `LabelEncoder` to convert them into numerical representations.

```
# Encode categorical features
```

```
# Encode categorical features
le = LabelEncoder()
categorical_cols = healthcare_df.select_dtypes(include='object').columns

for col in categorical_cols:
    healthcare_df[col + '_Encoded'] = le.fit_transform(healthcare_df[col])
    print(f"Original categories for '{col}': {le.classes_}")

print("\nDataFrame Head after encoding categorical variables:")
display(healthcare_df.head())
```

```
Original categories for 'Gender': ['Female' 'Male' nan]
Original categories for 'Medical Condition': ['Arthritis' 'Asthma' 'Cancer' 'Diabetes' 'Healthy' 'Hypertension'
 'Obesity' nan]
Original categories for 'random_notes': ['###' '??' 'ipsum' 'lorem']

DataFrame Head after encoding categorical variables:
```

	Age	Gender	Medical Condition	Glucose	Blood Pressure	BMI	Oxygen Saturation	LengthOfStay	Cholesterol	Triglycerides	...	Physical Activity	Diet Score	Family History	Stress Level	Sleep Hours	random_notes	noise_c
0	46.0	Male	Diabetes	137.04	135.27	28.90	96.04	6	231.88	210.58	...	-0.20	3.54	0	5.07	6.05	lorem	-137.0572
1	22.0	Male	Healthy	71.58	112.27	26.29	97.54	2	165.57	129.41	...	6.12	5.90	0	5.87	7.72	ipsum	-11.2306
2	50.0	NaN	Asthma	95.24	138.32	22.53	90.31	2	214.94	165.55	...	5.01	4.65	1	3.09	4.82	ipsum	98.3311
3	57.0	NaN	Obesity	NaN	130.53	38.47	96.60	6	197.71	182.13	...	3.16	3.37	0	3.01	6.33	lorem	44.1871
4	66.0	Female	Hypertension	95.15	178.17	31.12	94.90	4	259.53	115.85	...	3.56	3.40	0	6.38	6.64	lorem	44.8314

5 rows × 23 columns

Now, we will scale numerical features using `MinMaxScaler` to bring them into a standard range (0 to 1). We'll exclude the original categorical columns and the newly encoded ones, as well as `id`.

```python
# Scale numerical features using MinMaxScaler
scaler = MinMaxScaler()

# Identify numerical columns to scale (excluding original categorical and 'id')
# Also exclude newly created encoded columns from scaling
all_cols = set(healthcare_df.columns)
encoded_cols = [col for col in healthcare_df.columns if '_Encoded' in col]
original_categorical_cols = set(categorical_cols)

# Numerical columns are those that are not object type, not encoded, and not 'id'
numerical_cols = [col for col in healthcare_df.select_dtypes(include=np.number).columns
                  if col not in encoded_cols and col != 'id']

for col in numerical_cols:
    healthcare_df[col + '_Scaled'] = scaler.fit_transform(healthcare_df[[col]])

print("\nDataframe Head after scaling numerical features:")
display(healthcare_df.head())
```

```
Dataframe Head after scaling numerical features:
```

	Age	Gender	Medical Condition	Glucose	Blood Pressure	BMI	Oxygen Saturation	LengthOfStay	Cholesterol	Triglycerides	...	Triglycerides_Scaled	Hbalc_Scaled	Smoking_Scaled	Alcohol_Sc
0	46.0	Male	Diabetes	137.04	136.27	28.90	96.04	6	231.88	210.08	...	0.524877	0.476872	0.0	
1	22.0	Male	Healthy	71.58	113.27	26.29	97.54	2	166.57	129.41	...	0.342102	0.179515	0.0	
2	50.0	NaN	Asthma	95.24	138.32	22.53	90.31	2	214.94	165.35	...	0.423050	0.255507	0.0	

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Finally, we will split the dataset into training and testing sets. We'll use the `target` column (assuming 'heart_disease' is the target variable) as our dependent variable `y`, and all other relevant preprocessed features as our independent variables `X`.

```python
# Split dataset into training and testing sets

# Define the target variable (assuming 'LengthOfStay' is the intended target based on data content)
target_column = 'LengthOfStay'
y = healthcare_df[target_column]

# Identify feature columns for X
# Include all _Encoded categorical columns and _Scaled numerical columns,
# but exclude the scaled version of the target variable itself.
X_cols = [col for col in healthcare_df.columns if col.endswith('_Encoded') or col.endswith('_Scaled')]
if f'{target_column}_Scaled' in X_cols:
    X_cols.remove(f'{target_column}_Scaled')

X = healthcare_df[X_cols]

# Split the data, removing 'stratify=y' as LengthOfStay is likely a continuous target for regression.
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print(f"\nShape of X_train: {X_train.shape}")
print(f"Shape of X_test: {X_test.shape}")
print(f"Shape of y_train: {y_train.shape}")
print(f"Shape of y_test: {y_test.shape}")

print("\nFirst 5 rows of X_train:")
display(X_train.head())
```

```
Shape of X_train: (24000, 19)
```

Task 4 – Employee Salary Data Preprocessing

Task:

Clean and preprocess an employee salary dataset using AI scripts.

Instructions:

• Handle missing values

• Detect and remove salary outliers.

• Standardize department names (lowercase).

• Apply label encoding to job location, department.

Dataset link:

https://www.kaggle.com/code/vishantmathur/employee-salary

Expected Output:

• A structured dataset ready for HR analytics.